

FLOODLIGHT API EVALUATION

Port Statistics & Bandwidth Monitoring Mechanism

OVERVIEW

In the Floodlight OpenFlow controller ecosystem, registering for or requesting port statistics updates depends on whether the bandwidth monitoring application is implemented externally (via Northbound REST interface) or internally (as a native Java module).

1. EXTERNAL APPLICATION (NORTHBOUND REST API)

For external applications or standalone monitoring scripts, the developer uses Floodlight's **Statistics and Bandwidth REST API** endpoints provided by the Statistics Module.

- **Primary API Endpoint:** `/wm/statistics/bandwidth/<switch_id>/<port_id>/json` or `/wm/core/switch/<switch_id>/port/json`
- **Operation Mechanism:** Floodlight periodically polls connected OpenFlow switches using standard **OFPortStatsRequest** messages and processes the returned **OFPortStatsReply** messages. The developer's application queries these REST endpoints on a scheduled basis to retrieve calculated bit rates, byte counts, and packet counts.

2. INTERNAL MODULE (NATIVE JAVA API)

If the developer is writing an in-controller module in Java, they use Floodlight's internal service registry to bind to the **IStatisticsService** interface.

- **Primary Service Interface:** `net.floodlightcontroller.statistics.IStatisticsService`
- **Operation Mechanism:** The module requests port statistics directly from the **IStatisticsService** or registers listener callbacks with the core switch manager (**IOFSwitchListener**) to handle port status and counter update events.

SUMMARY CONCLUSION

To monitor port bandwidth and trigger threshold alerts, the developer would most likely use the **Bandwidth REST API** (`/wm/statistics/bandwidth/...`) for external polling applications, or integrate directly with the **IStatisticsService** Java interface for internal controller applications.